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# FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

## MICROPROCESSOR AND MICROCONTROLLERS

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

**1. a.** Write the Comparison between Microprocessor and Microcontroller. **05**

**b.** Write the Block Diagram of 8051 and Explain its features. **09**

**2. a.** Explain 5 Different Addressing modes with examples. **07**

**b.** What will be the content of the Accumulator after the execution of these instructions? **07**

(Assume initial value of the Accumulator= 0FFH and CY=0)

**i.** RR A

**ii.** RRC A

**iii.** RLC A

**iv.** ANL A, # 0F0H

**v.** ORL A, # 0F0H

**vi.** XRL A, # 0F0H

**3. a.** What is Stack? Explain PUSH and POP operations with an example. **07**

**b.** Show the contents of the Stack and Stack pointer for the following Instructions. **07**

```
MOV   SP, #5FH
MOV   R2, #25H
MOV   R1, #12H
MOV   R4, #0F3H
PUSH  2
PUSH  1
PUSH  4
```

**OR**

**4. a.** Write the different possible ways by which infinite loop can be created in ALP of 8051. Extend the concept to develop a code to continuously toggle the bits of Port1 with some delay. **07**

**b.** Write an Assembly level program to find the Largest of Ten given numbers. Assume suitable data. **07**

**5. a.** Illustrate with diagram the significance of bits in TMOD register to control timer/counter module in 8051 microcontroller. **07**

**b.** Explain how serial communication is done in 8051? **07**

**OR**

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| **6.** | **a.** | Explain the use of SCON and PCON registers. | **07** |
| | **b.** | Write an 8051 Assembly Language Program to toggle all the bits of P0 and P2 continuously with a 250ms delay. | **07** |
| **7.** | **a.** | Discuss the interrupts present in 8051. | **07** |
| | **b.** | Explain the use of IE and IP Registers. | **07** |

**OR**

| | | | |
|---|---|---|---|
| **8.** | **a.** | Discuss Interfacing of Switch and Relay with 8051. | **07** |
| | **b.** | Elaborate Interfacing of LCD with 8051 | **07** |